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CRN:

TECHNOLOGICAL UNIVERSITY DUBLIN
TALLAGHT CAMPUS

TU736 & TU936

BSc in Sports Science & Health

Year 1

SEMESTER 2
EXAMINATIONS 2023/24

Principles of Physics

Internal Examiner:
Dr Eugene Hickey

External Examiner:

Exam Duration:
Two hours in duration.

Instructions:
Please answer question one and any other two questions. A table of useful equations and constants is at the end of the exam paper.

Question 1 (Compulsory) (40 marks)

Answer any ten from the following fifteen parts. Each part is worth four marks

a) Express 0.00507J in scientific notation.

b) Convert 3400μm to mm.

c) Convert a density of 3.6g/mL to kg/m^3.

d) A golf putt on a flat green takes 6.2s to travel the 7.5m before stopping at the hole. Calculate the initial velocity of the ball.

e) What is the range of values for the coefficient of friction, μ?

f) What is the SI unit for power?

g) What are the three major areas of storage for carbohydrates in the body and give an indication of how much glycogen is in each.

h) An exercise regime uses 120kJ. Express this energy in Calories.

i) What is the crossover concept and how is influenced by endurance training?

j) After impact, a golf ball rotates at 2150RPM = 35.8Hz. Convert this frequency to angular velocity and calculate the maximum velocity of the surface of the golf ball. The golf ball radius is 2.135cm.

k) A torque of 2.8Nm produces an angular acceleration of 8.3rads/s^2. Calculate the moment of inertia of the body.

l) A golf ball (I_{ball} = 3.357x10^{-5}kgm^2) is spinning with rotational kinetic energy 0.7J Calculate its angular velocity.

m) Calculate the magnitude of the vector $\vec{A} = \vec{x} - 3\vec{y}$.

n) Calculate the dot product, $\vec{A}.\vec{B}$, of the two vectors $\vec{A} = \vec{x} - 3\vec{y} - \vec{z}$ and $\vec{B} = 2\vec{x} - \vec{y} + 2\vec{z}$.

o) Calculate the cross product, $\vec{A}x\vec{B}$, of the two vectors $\vec{A} = \vec{x} - 3\vec{y} - \vec{z}$ and $\vec{B} = 2\vec{x} - \vec{y} + 2\vec{z}$

Question 2 **(30 marks)**

a) What is meant by "*Momentum*"? (5 marks)

b) Why is momentum conservation important? (5 marks)

c) A golf club with a clubhead of 0.35kg strikes a golf ball of mass 0.046kg. The clubhead speed before impact is $52ms^{-1}$, after impact it is $41.9ms^{-1}$.

i) Show that the launch speed of the golf ball is $76.9ms^{-1}$. (6 marks)

ii) Calculate the kinetic energy of the clubhead before impact. (4 marks)

iii) Calculate the sum of the kinetic energy of the clubhead and the kinetic energy of the ball after impact. (4 marks)

iv) Calculate the loss in kinetic energy. (2 marks)

v) Calculate the Coefficient of Restitution of the golf club. (4 marks)

vi) For a golf shot, the *smash factor* is defined as the speed of the golf ball after impact divided by the speed of the golf club before impact. Calculate the smash factor for this golf shot. (4 marks)

Question 3 **(30 marks)**

a) What is meant by a ***torque***? (3 marks)

b) What are the SI units for torque? (3 marks)

c) A cyclist accelerates from 7m/s to 12m/s during a sprint. The acceleration takes 8s. The bicycle wheel radius is 0.622m and a mass of 0.8kg

i) Show that the acceleration of the bicycle during the 8s is $0.625 m/s^2$. (3 marks)

ii) Show that the bicycle travels 76m during the 8s. (3 marks)

iii) Show that the initial angular velocity of the bicycle wheel when it is travelling at 7m/s is 11.25rad/s. (3 marks)

iv) Show that the final angular velocity of the bicycle wheel when it is travelling at 12m/s is 19.3 rad/s. (3 marks)

v) Show that the angular acceleration of the bicycle wheel during the 8s is $1.0 rad/s^2$. (3 marks)

vi) Show that the moment of inertia of the bicycle wheel is $0.309 kgm^2$. (3 marks)

vii) Show that the necessary torque applied to the bicycle wheel to cause the angular acceleration is 0.309Nm. (3 marks)

viii) If the cycle pedal is 0.12m long, and assuming 90% of the cyclist's torque gets applied to the wheels, what is the force applied by the rider to the pedal? (3 marks)

Question 4 **(30 marks)**

a) Which of the following operations on two vectors produce an output vector?

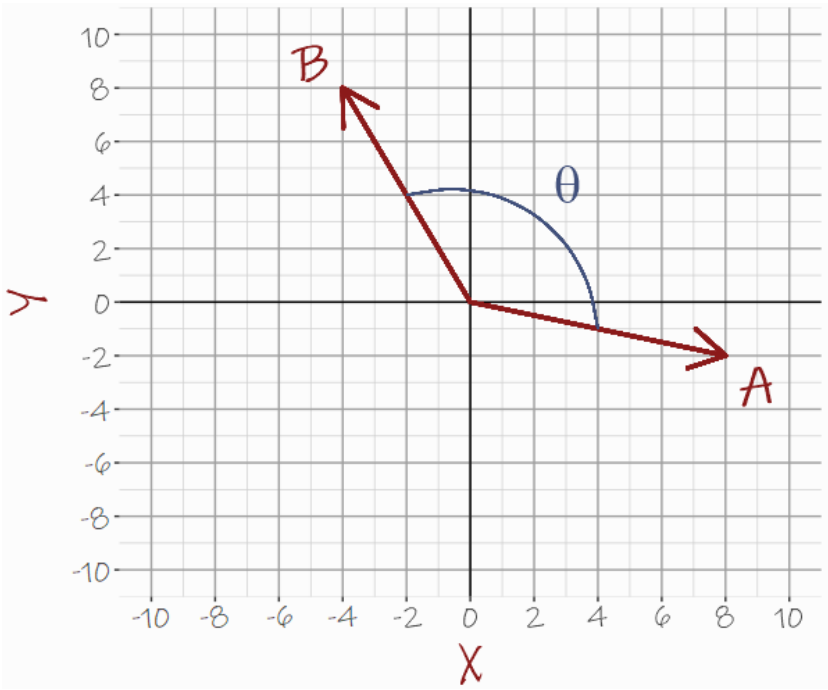
i) the dot product of two vectors, $\vec{A} \bullet \vec{B}$.

ii) the cross product of two vectors, $\vec{A} \times \vec{B}$.

iii) the magnitude of a vector, $|\vec{A}|$. (3 marks)

b) The angular momentum, $\vec{L}$, of an object moving with momentum, $\vec{p}$, on a circle of radius, $\vec{r}$, is given by , $\vec{L} = \vec{r} \times \vec{p}$. Draw a diagram showing $\vec{L}$, $\vec{r}$, and $\vec{p}$ using the right-hand rule to give their correct relative directions. (6 marks)

c) For the two vectors shown below:



i) Calculate their components, A_x, A_y, B_x, and B_y. (3 marks)

ii) Calculate their dot product, $\vec{A} \bullet \vec{B}$. (3 marks)

iii) Calculate the magnitude of $\vec{A}$ (3 marks)

iv) Calculate the magnitude of $\vec{B}$. (3 marks)

v) Calculate the cosine of the angle between $\vec{A}$ and $\vec{B}$. (3 marks)

vi) Calculate the angle, θ, between $\vec{A}$ and $\vec{B}$. (3 marks)

vii) Calculate the size and direction of the cross product, $\vec{A} \times \vec{B}$ (3 marks)

Useful Equations & Constants

Density = mass / volume

$v = v_0 + at$

$s = v_0 t + \frac{1}{2} at^2$

$v^2 = v_0^2 + 2as$

$s = (v + v_0)t/2$

$p = mv$

$W = mg$

$F_{fric} = \mu F_{reac} = \mu mg$

$P.E. = mgh$

$K.E. = \frac{1}{2} mv^2$

$$COR = \frac{v_{A_{final}} - v_{B_{final}}}{v_{A_{initial}} - v_{B_{initial}}}$$

$$COR = \sqrt{\frac{bounce\ height}{drop\ height}}$$

$\omega = 2\pi f$

$T = 1/f$

$v = \omega r$

$a = v\omega = \omega^2 r = v^2/r$

$T = F\,d$

$\omega = \omega_0 + \alpha t$

$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$

$\omega^2 = \omega_0^2 + 2\alpha\theta$

$\theta = (\omega + \omega_0)t/2$

$L = I\,\omega$

$T = I\alpha$

$I_{disc} = \frac{1}{2}MR^2$

$I_{bat} = \frac{1}{3}MR^2$

$I_{hollow\text{-}ball} = \frac{2}{3}MR^2$

$I_{solid\text{-}ball} = 2/5\ MR^2$

$RKE = \frac{1}{2}I\omega^2$

$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y$

$\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}| \cos\theta$

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\vec{x} + (A_z B_x - A_x B_z)\vec{y} + (A_x B_y - A_y B_x)\vec{z}$$

$g = 9.81 m/s^2$

1 Calorie = 4184J

Energy production: 21.4kJ / L of O_2